Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]

[18MSCH3H01]

THIRD SEMESTER M.Sc. UNIVERSITY EXAMINATION, NOVEMBER - 2019

INORGANIC CHEMISTRY - III

Time:03 Hours **Maximum Marks:70**

Instructions: 1. Question **1** is compulsory, carries **10** marks.
2. Questions **2 to 7** carry **15** marks each. Answer any Four.

1. Answer any FIVE of the following. **5X2 = 10**

a. List the factors favors the outer sphere reactions.

b. The stability constant for polydentate ligands is high, justify.

c. Define *ansa*-metallocene with examples.

d. What is σ-bond metathesis reaction? Give an example.

e. What is 14n + 2 rule for organometallics?

f. Explain photocatalytic CO_2 reduction.

2a. Describe the *trans* effect with suitable examples using square planar complexes.

2b. Discuss the Marçus theory of electron transfer with potential energy diagrams. **(10+5)**

3a. Explain the effect of macrocyclic ligands on the stability of metal complexes with a suitable example.

3b. Predict and explain the structure of the macrocyclic ligand, cryptand C222, and its complexes with M^+ (M = Li, Na, K and Cs) with correct order of formation constant.

3c. The following chemical reaction obeys associative mechanism:

$$[MX_6] + Y \cdot \longrightarrow [MX_5Y]$$

Explain the various steps involved in the reaction and deduce the rate law. **(5+5+5)**

4a. Write a note on the synthesis, structure and bonding in the following:

i. Transition metal complexes containing alkenes, and

ii. Transition metal complexes containing alkyne.

4b. Describe the structure and bonding in metal carbonyls. **(10+5)**

5a. Write notes on the following:

i. Collman's reagent

ii. Homocuprate,

iii. Heterocuprate and

iv. Lipshutz reagent

5b. Explain hydroformylation reaction with mechanism. **(10+5)**

6a. Using PSEP theory, interpret the possible structure for $[B_7H_7]^{2-}$, B_5H_9, B_4H_{10} and B_5H_{11}.

6b. Define metallacarboranes. Write the synthesis and structure of $[\eta^5\text{-}C_2B_9H_{11}FeCp]^-$ and $[Fe(\eta^5\text{-}C_2B_9H_{11})_2]^{2-}$. **(10+5)**

7a. Define photoredox catalysis and explain water splitting mechanism.

7b. Outline the photoredox reactions in cobalt and chromium complexes. **(5+10)**

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